

Plants, Part 2: Photosynthesis

Test · 30 marks total

Name: _____ Date: _____ Score: _____ / 30

Section A — Where photosynthesis happens

A1. Name the organelle where photosynthesis takes place.

[1 mark]

A2. Name the green pigment that absorbs light energy for photosynthesis.

[1 mark]

A3. Palisade mesophyll cells are found near the top of the leaf and are packed with chloroplasts. Explain how this adapts them for photosynthesis.

[2 marks]

Section B — The photosynthesis equation

B1. Write the word equation for photosynthesis.

[2 marks]

B2. State where the two reactants for photosynthesis come from.

[2 marks]

B3. Explain why photosynthesis is described as an endothermic reaction.

[2 marks]

B4. State three things a plant might do with the glucose it produces.

[3 marks]

Section C — Factors affecting the rate

C1. State three factors that affect the rate of photosynthesis.

[3 marks]

C2. Explain why increasing carbon dioxide concentration increases the rate of photosynthesis.

[2 marks]

C3. Explain why photosynthesis rate falls at very high temperatures, even though light and CO₂ are plentiful.

[2 marks]

C4. What name is given to the factor that is limiting the rate of photosynthesis at a given moment?

[1 mark]

C5. A student increases light intensity but the rate of photosynthesis does not increase. Suggest why.

[2 marks]

Section D — Translocation, nutrition & disease

D1. What is translocation?

[2 marks]

D2. Name the transport tissue involved in translocation.

[1 mark]

D3. A plant has yellow leaves but is growing at a normal rate. Suggest which mineral ion it is short of, and explain the symptom.

[2 marks]

D4. A plant is stunted (small) for its age. Suggest which mineral ion it is short of, and explain why this causes the symptom.

[2 marks]

D5. Give three visible signs that a plant might be diseased.

[2 marks]